Exercise Sheet: Monte Carlo Simulation and Variance Reduction

Objective

In this exercise, you will apply Monte Carlo simulation techniques to estimate the expected carbon footprint of training a machine learning model under uncertainty. You will also implement variance reduction methods to improve estimation efficiency.

Background

Training machine learning models consumes energy, leading to carbon emissions. However, the total emissions are uncertain due to variability in:

- Training time
- Hardware power consumption
- Carbon intensity of electricity

We model total emissions as:

$$E = T \cdot P \cdot C$$

where:

- $T =$ training time (hours)
- $P =$ power usage (kW)
- $C =$ carbon intensity (kg CO_2/kWh)

Exercise 1: Basic Monte Carlo Simulation

Assume the following distributions:

- $T \sim \text{LogNormal}(\mu_T, \sigma_T)$
- $P \sim \mathcal{N}(\mu_P, \sigma_P)$
- $C \sim \text{Uniform}(a, b)$

Tasks

1. Implement a Monte Carlo estimator for $\mathbb{E}[E]$ using n samples.
2. Compute:
    - Sample mean
    - Sample variance
    - Standard error

3. Construct a 95% confidence interval.
4. Analyze convergence by plotting the estimate as a function of n.

Discussion:

- How does the estimation error change with increasing n?
- Relate your observations to the Law of Large Numbers and Central Limit Theorem.

Exercise 2: Importance Sampling

Rare but high-impact events (e.g., high carbon intensity) can dominate emissions.

Tasks

1. Modify the sampling of C using a proposal distribution $q(C)$ that over-samples high values.
2. Compute importance weights:
$$w_i = \frac{p(C_i)}{q(C_i)}$$
3. Implement the importance sampling estimator:
$$\hat{E} = \frac{1}{n}\sum_{i=1}^{n} T_i \cdot P_i \cdot C_i \cdot w_i$$
4. Compare results with standard Monte Carlo:
   - Mean estimate
   - Variance
   - Convergence speed

Discussion:

- Why does importance sampling help in this scenario?
- What happens if the proposal distribution is poorly chosen?

Exercise 3: Control Variates

We define a control variable:
$$X = T \cdot P$$
Assume that $\mathbb{E}[X] = \mathbb{E}[T] \cdot \mathbb{E}[P]$ is known analytically.

Tasks

1. Estimate the covariance $\mathsf{Cov}(X, E)$ and variance $\mathsf{Var}(X)$.
2. Compute the optimal coefficient:
$$c^* = \frac{\mathsf{Cov}(X, E)}{\mathsf{Var}(X)}$$
3. Construct the control variate estimator:
$$E_{\mathsf{CV}} = E - c^*(X - \mathbb{E}[X])$$

4. Estimate $\mathbb{E}[E]$ using the adjusted samples.
5. Compare variance with the standard Monte Carlo estimator.

Discussion:

- Why is X a good control variable?
- Under what conditions does this method work best?

Final Questions

1. Compare all three methods:
   - Standard Monte Carlo
   - Importance Sampling
   - Control Variates
2. Which method achieved the lowest variance? Why?
3. How can these techniques help design more sustainable AI systems?